

Project Proposal (Proposed Solution) Template

Phase: Project Initialization and Planning Phase

Project Title: Food Consumer Behaviour Analysis

Team ID: PNT2022TMID01234

Date: 15 October 2024

Maximum Marks: 3 Marks

Project Proposal Summary

This project proposal outlines a solution to address a specific problem. With a clear objective, defined scope, and a concise problem statement, the proposed solution details the approach, key features, and resource requirements, including hardware, software, and personnel.

Project Overview & Proposed Solution Details

Section	Description / Details
Project Overview	
Objective	Develop an end-to-end interactive Business Intelligence and Data Analytics solution investigating Food Consumer Behavior across demographic segments, ordering channels, dining frequencies, and spending elasticity to provide actionable insights for restaurants, aggregators, and food brands.
Scope	The project encompasses data ingestion of multi-dimensional food consumer dining and transactional records, rigorous data cleansing, exploratory data analysis, interactive dashboard engineering in Tableau Public, executive story point design, and deployment within a modern responsive web portal.
Problem Statement	
Description	Food service providers face severe customer churn, erratic demand forecasting, and suboptimal menu pricing due to fragmented data across dine-in and digital channels, lack of granular demographic visibility, and poor insight into price sensitivity drivers.
Impact	Solving this problem empowers restaurants and food platforms to optimize menu pricing, personalize promotional discounts, streamline kitchen inventory prep based on temporal surges, and boost customer retention and profitability by 20-35%.
Proposed Solution	
Approach	Data Collection & Preprocessing: Clean, normalize, and validate consumer survey and transactional data using Python (Pandas/NumPy). Exploratory Data Analysis: Perform statistical correlation and demographic cohort segmentation. Tableau Visual Engineering: Build 8+ KPI-driven visualizations, responsive dashboards, and narrative story points. Web Deployment: Embed live interactive visualizations inside a responsive, modern React/Vite application.
Key Features	- Multidimensional demographic cohort filtering (Age, Income, Family Size) - Channel dynamics breakdown (Dine-in, Food Delivery Apps, Takeout) - Spending elasticity and Average Order Value (AOV) correlation models - Peak temporal ordering heatmaps (Month, Day-of-Week, Hour) - Interactive dual-mode portal (Live Dashboard & Guided Executive Story)

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	Intel Core i5/i7 or AMD Ryzen 5/7 (6+ Cores), 2.5 GHz+ base clock

Resource Type	Description	Specification/Allocation
Memory	RAM specifications	16 GB DDR4/DDR5 RAM for smooth data preprocessing and rendering
Storage	Disk space for data, models, and logs	512 GB SSD (NVMe high-speed storage)
Software		
Frameworks	Python & Frontend frameworks	React 19, Vite 8, Lucide React (Web UI)
Libraries	Data science & visualization libraries	Pandas, NumPy, Matplotlib, Seaborn, Tableau Public Engine
Development Environment	IDE, version control & tools	Visual Studio Code, Tableau Desktop / Public, Git, GitHub
Data		
Data	Source, size, format	Food Consumer Behavior Survey & Orders Dataset (~10,000+ records, CSV format, 15+ attributes)